UNIVERSITY OF MAURITIUS

SIS 3204Y(5) - NATURAL LANGUAGE PROCESSING

Week 5 - Additional Sample Exam Questions

Questions 11-20 | Sample Answers and Marking Rubrics

This document contains 10 additional Week 5 revision questions designed to complement Questions 1-10, with emphasis on hidden states, embeddings, information flow, gate interpretation, recurrent architectures and the transition to attention and Transformers.

Question 11 - 6 marks

Explain the role of the hidden state in an RNN. Why can it be considered a form of memory?

Sample Answer

The hidden state stores a representation of information obtained from the sequence up to the current time step. At each step, the RNN combines the current input with the previous hidden state to calculate a new hidden state. The new state is then passed to the next time step. It can therefore be considered a form of memory because information from earlier words can influence the processing of later words.

Marking Rubric

Expected element	Marks
Correctly defines hidden state	2
Explains use of previous hidden state and current input	2
Explains passing the state forward	1
Connects this mechanism to memory/context	1
Total	**6**

Question 12 - 6 marks

Explain how word embeddings are used as inputs to an RNN for an NLP task.

Sample Answer

An RNN cannot directly process words as textual symbols. Each word is therefore converted into a numerical vector using a word embedding. For a sentence, the embedding vectors are presented to the RNN sequentially. At each time step, the current embedding is combined with the previous hidden state to generate a new hidden state. In this way, the RNN processes both the numerical representation of the current word and contextual information from previous words.

Marking Rubric

Expected element	Marks
Identifies need for numerical representation	1
Explains word embeddings	1
Embeddings supplied sequentially	1
Current embedding combined with previous hidden state	2
Links process to contextual representation	1
Total	**6**

Question 13 - 8 marks

Describe how an RNN can be used to classify a sentence into one of three categories: positive, neutral or negative.

Sample Answer

The sentence is first tokenised and each token is represented using an embedding. The embeddings are then processed sequentially by the RNN. At each time step, the hidden state is updated using the current word embedding and the previous hidden state. After the sequence has been processed, a representation such as the final hidden state can be passed to an output layer. For three-class sentiment classification, a softmax layer can produce probabilities for positive, neutral and negative. The class with the highest predicted probability can then be selected as the model's prediction.

Marking Rubric

Expected element	Marks
Tokenisation/preparation	1
Embeddings	1
Sequential RNN processing	1
Hidden-state updates	2
Final representation used for classification	1
Softmax/three probabilities	1
Final class prediction	1
Total	**8**

Question 14 - 8 marks

Describe how information flows through an LSTM cell at one time step.

Sample Answer

At a time step, the LSTM receives the current input, the previous hidden state and the previous cell state. The forget gate determines how much information from the previous cell state should be retained. The input gate controls how much new candidate information should be added. These operations produce the updated cell state. The output gate then controls how much information from the updated cell state is exposed through the new hidden state. The new cell state and hidden state are passed to the next time step.

Marking Rubric

Expected element	Marks
Identifies current input	1
Previous hidden/cell states	1
Forget gate	1
Input gate	1
Updated cell state	1
Output gate	1
New hidden state	1
States passed to next time step	1
Total	**8**

Question 15 - 6 marks

Consider an LSTM processing a long sentence. Explain what it means conceptually if the forget gate produces a value close to 0 for some stored information and what it means if it produces a value close to 1.

Sample Answer

The forget gate determines how much previous cell-state information should be retained. A value close to 0 means that very little of the corresponding previous information is retained, so the LSTM effectively forgets it. A value close to 1 means that most of that information is preserved in the cell state. This mechanism allows the LSTM to remove information that is no longer useful while retaining information that may be important later in the sequence.

Marking Rubric

Expected element	Marks
Identifies purpose of forget gate	1
Interprets value near 0	2
Interprets value near 1	2
Connects behaviour to selective memory	1
Total	**6**

Question 16 - 8 marks

A GRU is processing the sentence “The food was not very good.” Explain how its hidden state and gates can help it build a representation of the sentence.

Sample Answer

Each word is converted into an embedding and processed sequentially. The hidden state carries contextual information from one word to the next. The reset gate determines how much previous information should contribute when new candidate information is generated, while the update gate determines how much of the previous hidden state should be retained relative to the new candidate information. As words such as “not” and “good” are processed, the hidden state is updated to reflect their relationship in context. The final hidden state can then be used for a task such as sentiment classification.

Marking Rubric

Expected element	Marks
Embeddings/sequential processing	1
Hidden state carries context	2
Reset gate correctly explained	1
Update gate correctly explained	2
Recognises contextual relationship such as 'not ... good'	1
Connects final state to prediction	1
Total	**8**

Question 17 - 8 marks

Explain two similarities and three differences between an LSTM and a GRU.

Sample Answer

Both LSTM and GRU are gated recurrent architectures designed to improve the modelling of dependencies in sequential data. Both control the flow of previous and new information rather than relying only on a simple recurrent hidden state. An LSTM uses three main gates - forget, input and output - whereas a GRU uses two main gates - reset and update. An LSTM maintains a separate cell state and hidden state, while a GRU does not maintain a separate cell state. Consequently, the GRU has a simpler architecture and generally fewer parameters than an LSTM.

Marking Rubric

Expected element	Marks
First valid similarity	1
Second valid similarity	1
Difference in number/type of gates	2
Cell-state difference	2
Complexity/parameter difference	2
Total	**8**

Question 18 - 8 marks

Distinguish between many-to-one and many-to-many recurrent neural network architectures. Give an NLP application for each.

Sample Answer

In a many-to-one architecture, several input elements are processed as a sequence to produce one output. Sentiment classification is an example: a sequence of words is processed and one sentiment label is produced for the complete text. In a many-to-many architecture, a sequence of inputs is associated with a sequence of outputs. Sequence labelling is an example. In Named Entity Recognition, for instance, individual tokens can receive labels such as PERSON, LOCATION or ORGANISATION.

Marking Rubric

Expected element	Marks
Correctly explains many-to-one	2
Appropriate many-to-one NLP example	2
Correctly explains many-to-many	2
Appropriate many-to-many NLP example	2
Total	**8**

Question 19 - 10 marks

Two teams are developing NLP systems. Team A uses a vanilla RNN for short text sequences. Team B uses an LSTM for considerably longer sequences in which information appearing early may affect interpretation much later. Explain why these choices may be reasonable and discuss their limitations.

Sample Answer

Team A's choice may be reasonable because a vanilla RNN has a relatively simple architecture and can model sequential information using its hidden state. For short sequences, long-term dependency problems may be less important. However, the RNN can still suffer from vanishing gradients and may become unsuitable as sequence length and dependency distance increase. Team B's LSTM provides a cell state and forget, input and output gates. These mechanisms allow important information to be retained selectively over longer sequences and therefore address some of the limitations of the vanilla RNN. However, the LSTM has more parameters and greater computational complexity. The suitability of either architecture should ultimately be evaluated on the actual task and data.

Marking Rubric

Expected element	Marks
Explains why RNN may suit short sequences	2
RNN limitation	2
Explains why LSTM may suit longer dependencies	2
Relates gates/cell state to memory	2
LSTM complexity limitation	1
Appropriate task/data-based conclusion	1
Total	**10**

Question 20 - 10 marks

Explain the progression from standard neural networks to RNN to LSTM/GRU to Attention/Transformers in NLP. For each transition, identify the main limitation that motivated the next development.

Sample Answer

Standard feed-forward neural networks do not maintain a memory of previous inputs, which is a limitation for sequential text. RNNs address this by introducing recurrent hidden states so previous information can influence later processing. Vanilla RNNs, however, struggle with long-term dependencies, particularly because of problems such as vanishing gradients. LSTM and GRU introduce gating mechanisms that provide better control over memory and information flow. Although LSTM and GRU improve long-term memory, they remain recurrent architectures and process sequences sequentially. This limits parallelisation and can make processing long sequences inefficient. Attention provides more direct access to relevant information at different positions in a sequence. Transformers build on self-attention and avoid recurrent token-by-token processing, enabling greater parallelisation and scalability.

Marking Rubric

Expected element	Marks
Feed-forward NN limitation	1
Explains RNN solution	1
Identifies RNN long-term dependency problem	2
Explains LSTM/GRU gating solution	2
Identifies remaining sequential-processing limitation	1
Explains attention concept	1
Explains Transformer/self-attention and parallelisation	2
Total	**10**